APPENDIX
PREVIOUS YEAR QUESTIONS

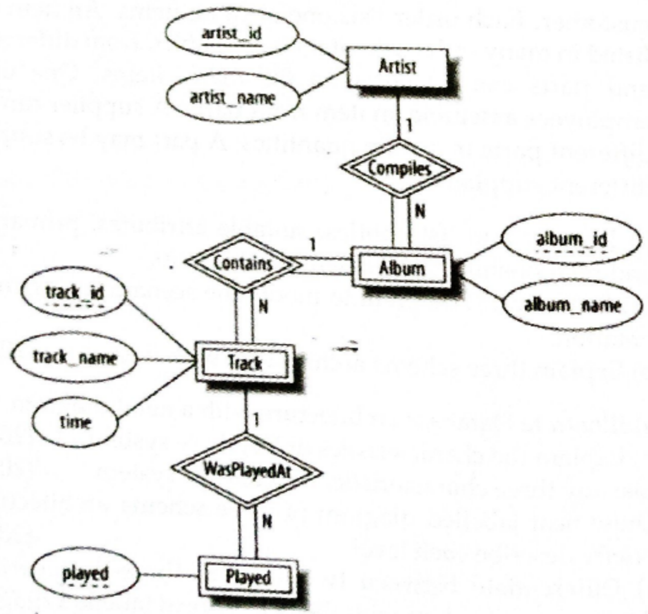
MODULE-1

1. List any three categories of database users, highlighting any one important characteristic of each category. (2021)
2. What are the major differences between structured, unstructured and semi structured data? (2021)
3. What is entity integrity? Why is it important? (2021)
4. Distinguish between Super key, Candidate key, and Primary key using a real convincing example (2021)
5. a) A company has the following scenario: There are a set of salespersons. Some of them manage other salespersons. However, a salesperson cannot have more than one manager. A salesperson can be an agent, for many customers. A customer is managed by exactly one salesperson. A customer can place any number of orders. An order can be placed by exactly one customer. Each order lists one or more items. An item may be listed in many orders. An item is assembled from different parts and parts can be common for many items. One or more employees assemble an item from parts. A supplier can supply different parts in certain quantities. A part may be supplied by different suppliers.

   (i) Identify and list entities, suitable attributes, primary keys, and relationships to represent the scenario.
   (ii) Draw an ER diagram to model the scenario using min-max notation.
   b) Explain three schema architecture with figure (2021)
6. a) Illustrate Database architecture with a ne-at diagram
   b) Explain the characteristics of Database system (2021)
7. List any three characteristics of database system (2022)
8. Draw neat labelled diagram of three schema architecture and briefly describe each level. (2022)
9. a) Differentiate between two-tier and three-tier client-server database architecture with the help of neat labelled diagrams.
   b) Draw an ER diagram based on the following information,

- Manufacturers have a name, which we may assume is unique, an address, and a phone number
- Products have a model number and a type. Each product is made by one manufacturer, and different manufacturers may have different products with the same model number. However, you may assume that no manufacturer would have two products with the same model number
- Customers are identified by their unique social security number. They have email addresses, and physical addresses. Several customers may live at the same (physical) address, but we assume that no two customers have the same email address
- An order has a unique order number, and a date. An order is placed by one customer. For each order, there are one or more products ordered, and there is a quantity for each product on the order. (2022)

10. a) Write briefly about any three types of database end users
 b) Interpret the following ER diagram (2022)



11. List any SIX major advantages of using a DBMS (2023)

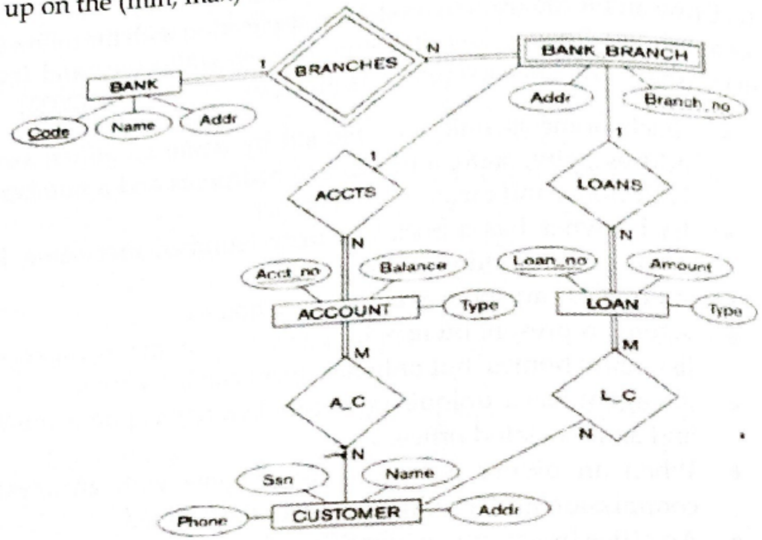
12. What is the concept of a weak entity used in data modelling? Define the terms owner entity type, Identifying relationship type. (2023)
13. Define primary key, candidate key and super key. (2023)
14. Draw an ER diagram to model the application with the following assumptions. Specify key attributes of each entity type and (min, max) constraints on each relationship type. (2023)

- Each home is uniquely defined by home identifier, street address, city, state, a number of bedrooms and a number of bathrooms and an associated owner.
- Each owner has a Social Security Number, first name, last name, phone, and profession.
- An owner can spouse one or more homes.
- Agents represent owners in the sale of a home. An agent can list many homes, but only one agent can list a home.
- An agent has a unique agent number, name, phone number and an associated office.
- When an owner agrees to list a home with an agent, a commission and a selling price are determined.
- An office has an office identifier, phone number, the manager name, address and an optional agent number.
- Many agents can work at one office.
- A buyer entity type has a Social Security Number, first name, last name, phone, preferences for the number of bedrooms and bathrooms, and a price range.
- An agent can work with many buyers, but a buyer works with only one agent.

15. List out any three database users and their functionalities. (2024)
16. Distinguish between schema and instance. (2024)
17. Consider the bank database given above and answer the following questions (2023)

 i. List the strong (non-weak) entity types in the ER diagram.
 ii. Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
 iii. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
 iv. List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type.

v. Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1,000 loans. How does this show up on the (min, max) constraints?



18. Differentiate between Super key, Candidate key and Primary key. (2024)

19. a) Explain the seven drawbacks of file processing system, compared to database approach. (2024)

b) Represent a "Library Management Software" using an ER diagram.

20. a) Explain the concept of three schema architecture with the help of a neat labelled diagram. (2024)

b) Represent an "Online Shopping Portal" using an ER diagram.

21.Compare weak entity and strong entity with examples. (2025)

22. What are the major differences between structured, unstructured and semi-structured data? (2025)

23. Define primary key, candidate key and super key. (2025)

24.a) With the help of a neat diagram explain the three schema architecture of DBMS (2025)

b) What is an attribute? Explain different types of attributes in the data base management system.

25. a) Draw the ER model of a company by considering the following constraints (2025)

- In a company, an employee works on many projects which are controlled by one department.
  - One employee supervises many employees.
  - An employee has one or more dependents.
  - One employee manages one department.

b) List different database users and explain each with examples.

MODULE-2

1. Illustrate the concept of trigger in SQL with an example. (2021)
2. Compare DDL and DML with the help of an example. (2021)
3. IIllustrate different anomalies in designing a database. (2021)
4. a) Study the tables given below and write relational algebra expressions for the queries that follow.

STUDENT(ROLLNO, NAME, AGE, GENDER, ADDRESS, ADVISOR)
COURSE(COURSEID, CNAME, CREDITS)
PROFESSOR(PROFID,PNAME, PHONE)
ENROLLMENT(ROLLNO, COURSEID, GRADE)

Primary keys are underlined. ADVISOR is a foreign key referring to PROFESSOR table. ROLLNO and COURSEID in ENROLLMENT are also foreign keys referring to the primary keys with the same name.

(i) Names of female students
(ii) Names of male students along with adviser name
(iii) Roll Number and name of students who have not enrolled for any course.

b) Explain the left outer join, right outer join, full outer join operations with examples (2021)

5. a) Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON(Ssn, Name, StartYear, DeptNo)
TRIP(Ssn, FromCity, Tocity, DepartureDate, RetumDate, TripId)
EXPENSE(TripId, AccountNo, Amount)

i) A trip can be charged to one or more accounts. Specify the foreign keys for this schema, stating any assumptions you make.

ii) Write relation algebra expression to get the details of salespersons who have travelled between Mumbai and Delhi and the travel expense is greater that Rs. 50000.
iii) Write relation algebra expression to get the details of salesperson who had incurred the greatest travel expenses among all travels made.
b) List the basic data types available for defining attributes in SQL? (2021)

6. Write short notes on Nested queries (2021)
7. Write briefly about any three relational database integrity constraints. (2022)
8. Differentiate between theta join and natural join operations. (2022)

9. a) Consider the following schema, (2022)
Suppliers (sid, sname, address)
Parts (pid. pname, color)
Catalog (sid, pid,cost)
The primary key fields are underlined.
Write relational algebra expressions for the following queries:
b) Find the name of parts supplied by supplier with sid=105

ii) Find the names of suppliers supplying some green part for less than Rs 1000
iii) Find the IDs of suppliers who supply some red or green part
iv) Find the names of suppliers who supply some red part

b) Differentiate between the following SQL statements
i) DROP and DELETE
ii)ALTER and UPDATE

10. a) Write SQL DDL statements based on the following database schema (Assume suitable domain types): (2022)

Employee (eid, name, designation, salary, comp_id)
Company (comp_id, cname, address, turnover)

b) Create the above mentioned tables assuming each company has many employees. Mention the primary key, foreign key and not null constraints.

i) Insert values into both the tables. Mention in which order insertions will be carried out.

ii) Modify the table Employee to include a new column "years-of-exp"
iv) Increment the salary of employees whose salary is less than Rs25000 by 5%.
b) Illustrate any three ways of using INSERT'statement in SQL.

11. a) For the relation schema below, give an expression in SQL for each of the queries that follows:

employee (ID,person_name, street, city) (2022)
works (ID ,company_name, salary)
company (company_name, city)
manages (ID ,manager_id)

b) Find the employees whose name starts with 'C'

ii) Find the name of managers of each company
iii) Find the ID, name, and city of residence of employees who works for "First Bank Corporation" and earns more than Rs50000
iv) Find the name of companies; whose employees earn a higher salary, on average,than the average salary at "First Bank Corporation"

12. What is the difference between the WHERE and HAVING clause? Illustrate with an example. (2023)
13. a) Consider the UNIVERSITY database with the following relations: (2023)

STUDENT (rollNo, name, degree, year, sex, deptNo, advisor)
DEPARTMENT (deptId, name, hod, phone)
PROFESSOR (empId , name, sex, startYear, deptNo, phone)
COURSE (courseId , cname, credits, deptNo)
ENROLLMENT(rollNo, courseId,sem,year, grade)
TEACHING(empId,courseId,sem,year ,classRoom)
PREREQUISITE (preRegCourse, courseID)

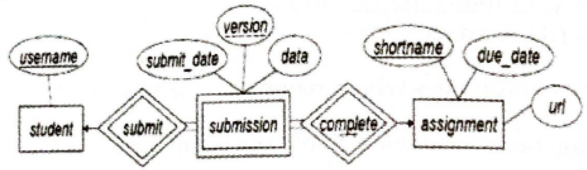
Write relational algebra expressions for the following queries:

i. For each department, find its name and the name, sex and phone number of the head of the department.
ii. Find courses offered by each department.

iii. Find those students who have registered for all courses offered in the department of Computer Science.
iv. Obtain the department Ids for departments with no lady professor.
v. Obtain the rollNo of girl students who have obtained at least one S grade.

b) What is a foreign key constraint? Why are such constraints important? What is referential integrity?

14. a) Convert the following ER diagram into a relational schema



b) Consider the following relation schema with referential integrity constraints:

STUDENT (<u>rollNo</u>, name, degree, year, sex, deptNo, advisor)

DEPARTMENT (<u>deptId</u>, name, hod, phone)

PROFESSOR (<u>empId</u>, name, sex, startYear, deptNo, phone)

Write SQL DDL statements for the following:
i. Create table STUDENT, DEPARTMENT, PROFESSOR including primary and foreign key integrity constraints.
ii. Add an address attribute in the table STUDENT
iii. Write an SQL statement to delete the "CS" department. Given the referential integrity constraints, explain what happens when this statement is executed.

15. a) What is an assertion? How they differ from triggers? (2023)
b) Consider the following relation schema and write SQL queries to find:

EMPLOYEE(Fname, Minit, Lname, <u>SSN</u>, Bdate, Address, Sex, Salary SuperSSN, Dno)
DEPARTMENT(Dname, <u>Dnumber</u>, MgrSSN, MgrStartDate)
DEPT_LOCATIONS (<u>Dnumber, Dlocations)</u>
PROJECT(Pname, <u>Pnumber</u>, Plocation,Dnum)
WORKS_ON(<u>ESSN,Pno</u>,Hours)

i. Retrieve the name and address of all employees who work for the 'Research' department.
ii. For each employee, retrieve the employee's name, and the name of his or her immediate supervisor.
iii. Retrieve the name of each employee who works on all the projects controlled by department number 5.
iv.Make a list of all project numbers for projects that involve an employee whose last name is 'Smith' as a worker or as a manager of the department that controls the project.
v. Retrieve the SSN of all employees who work on project number 1,2, or 3.

16. Explain the intersection operation in relational algebra with an example. (2024)
17. Outline concept of assertion in SQL with an example. (2024)
18. State insertion anomaly with suitable example. (2024)
19. a) Explain the left outer join, right outer join and full outer join operations with suitable examples. (2024)
b) Consider the following schema.

Suppliers(<u>sid</u> , sname, address)
Parts(<u>pid</u> , pname, color)
Catalog(<u>sid, pid</u> , cost)

The primary key fields are underlined; Foreign keys have the same name as primary keys. Assume integer domain for the attributes sid, pid, cost and string domain for the attributes sname, address, pname, color. Write relational algebra expressions for the given questions. (Use * symbol for natural join and ⋈ symbol for join)

(i) Find the names of suppliers who supply some red part.
(ii) Find the sids of suppliers who supply some red or green part.
(iii) Find the sids of suppliers who supply every red part or supply every green part.
(iv) Find the pids of parts supplied by at least two different suppliers.

20. a) What is constraint? Discuss about domain constraint, entity integrity and referential integrity constraint with suitable example. (2024)

b) Illustrate with an example the different steps involved in synthesizing an ER diagram into a relational schema.

21. Describe any three aggregate functions in SQL with example. (2024)

22. Consider the following relations: (2024)
Employee (<u>Employee-Id</u> , Employee-Name, Salary, Department-No)
Department (<u>Department-No</u> , Department-Name)

The primary key fields are underlined. Foreign keys have the same name as primary keys.

Write SQL queries for the following:

(i) Retrieve the employee names and their department names.
(ii) Retrieve department names and the average salary given by them.
(iii) Retrieve the ids of employees getting salary greater than the average salary of their department. (iv) For each department that has more than 4 employees, retrieve the department-No and the number of employees getting salary more than Rs. 50000.

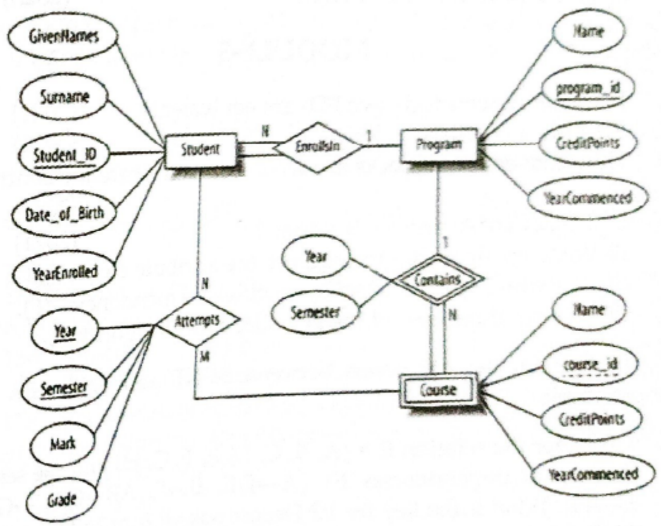
23. Compare SELECT and PROJECT operations on relational database with examples. (2025)
24. Differentiate between EQUI-JOIN and NATURAL JOIN (2025)
25. What is an assertion? How they differ from triggers? (2025)
26. a) An Employee relation has attributes: Employee-Id (numeric type), Name (character type), Salary (numeric type) and Dep-No (numeric type). (2025)
A Department relation has attributes: Department-Number (numeric type), Department-Name (character type), Dep-Manager-Id (numeric type).

Employee-Id is the primary key of Employee relation. Department-Number is the primary key of the Department relation. Dep-No attribute of Employee relation refers to the Department-Number attribute of Department relation and Dep Manager-Id attribute of Department relation refers to the Employee-Id attribute of Employee relation.

(i) Write create table statements by specifying necessary integrity constraints for creating these two relations in SQL.
(ii) Write SQL statement to insert the details of an employee John with id 101 with salary 5000 and working in department number 5.
(iii) Insert the details of a Research Department with Department Number 1 and it has not been assigned any manager.
(iv) Assume that a department with employees working in it is to be deleted. Specify the two options to manage this scenario.

b) What is meant by constrains in relational database? Explain each with example.

27. a) Write the rules for converting ER diagram to relational model and convert the following ER diagram to relational model.



b) Differentiate between the given below SQL statements with examples (2025)

(i) DROP and DELETE
(ii) ALTER and UPDATE

28. For the relation schema below, By assuming necessary key and referential integrity constraints, give an expression in SQL for each of the queries that follows: (2025)

employee (ID, person name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)

(i) Find the employees whose name starts with `C'
(ii) Find the name of managers of each company
(iii) Find the ID, name, and city of residence of employees who works for "First Bank Corporation" and earns more than Rs. 50000
(iv) Find the name of companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation"

29. Define views in SQL. Write simple SQL queries to create a view, update a view and drop a view. (2025)

MODULE-3

1. How can we conclude two FDs are equivalent? (2021)
2. Illustrate two phase locking (2021)
3. How conversions of locks are achieved in concurrency control? (2021)
4. a) i) What are Armstrong's axioms? (2021)
ii) Write an algorithm to compute the attribute closure of a set of attributes (X) under a set of functional dependencies (F).
iii) Explain three uses of the attribute closure algorithm.

b) Explain the difference between BCNF and 3NF with an example

5. Consider the relation R = {A, B, C, D, E, F, G, H} and the set of functional dependencies F= {A→DE, B→F, AB→C, C→GH, G→H}. What is the key for R? Decompose R into 2NF and then 3NF relations. (2021)
6. a) Explain the concepts behind the following: (2021)
i) Log-Based Recovery
ii) Deferred Database Modification.

b) Why is recovery needed in transaction processing?

7. a) Differentiate serial and concurrent schedules. Elaborate conflict serializability with suitable examples. (2021)

b) What are the desirable properties of transactions? Explain

8. Define Boyce Codd normal form. How does it differ from 3NF? (2022)
9. Suppose, a relational schema R (P,Q, R, S) and set of functional dependencies F and G are as follow:
F : { P→ Q, Q → R →S } G : { P'→QR, R →S }. Check the equivalency of functional dependencies F and G. (2022)
10. Write briefly on log based recovery (2022)
11. a) Consider a relation R(A, B, C, D, E) with FDs AB→C, AC→B, BC→A,D→E. Determine all the keys of relation R. Also decompose the relation into collections of relations that are in BCNF. (2022)
b) Write briefly on the different types of anomalies in designing a database.

12. a) Explain briefly the ACID properties of transaction. (2022)
b) Check whether the given schedules are conflict serializable or not

i) Sl : $R_1(X)$, $R_2(X)$, $R_1(Y)$, $R_2(Y)$, $R_3(Y)$, $W_1(X)$, $W_2(Y)$

ii)S2 : $R_1(X)$, $R_2(X)$, $R_2(Y)$, $W_2(Y)$, $R_1(Y)$, $W_1(X)$

13. What is a two phase locking protocol? How does it guarantee serializability? (2022)
14. Define the term functional dependency. Why are some functional dependencies called trivial? (2023)
15. List Armstrong Axiom rules (2023)
16. List the ACID properties of transactions. (2023)
17. Consider a relation R with five attributes (A,B,C,D,E). You are given the following dependencies: A → B, BC→E, and ED →A.
i. List all keys for R.
ii. Is R in 3NF?
iii. Is R in BCNF? (2023)
18. Explain with examples 2NF, 3NF and BCNF (2023)
19. a) What is a schedule? Define the concepts of recoverable, cascade less and strict schedules and compare them in terms of their recoverability. (2023)
b) Which of the following schedule is conflict serializable? For each serializable schedule determine the equivalent serial schedule.

(a) r1(X);r3 (X); w1 (X); r2(X); w3(X)

(b) r1(X);r3 (X); w1 (X); r2(X)
(c) r3(X);r2 (X); w3 (X); r1(X); w1(X)

20. What is the two-phase locking (2PL) protocol? How does it guarantee serializability. How strict 2PL differs from basic 2PL? (2023)
21. State the Armstrong's axioms of FD. (2024)
22. Explain the different properties of a transaction. (2024)
23. Define lNF,2NF,3NF and BCNF with suitable examples. (2024)
24. a) Describe the different states of a transaction with the help of a neat sketch. (2024)
    b) What is defined database modification? How it is different from immediate database modification? Explain the recovery steps in deferred database modification with an example.
25. a) Determine if the following schedule is recoverable. Is the schedule cascadeless? Justify your answer. (2024)
    rl(X),r2(Z),rl(Z), r3(X), r3(Y), w1(X), cl, w3(Y), c3, r2(Y), w2(Z),w2(Y),c2 (Note: ri(X)/wi(X) means transaction Ti issues read/write on item X; ci means transaction Ti commits.)

    b) Check whether the following schedule is conflict serializable or not and find an equivalent serial schedule if possible. r1(X),r2(Z),r1(Z),r3(X),r3(Y),w1(X),w3(Y),r2(Y),w2(Z),w2(Y) (Note: ri(X)/wi(X) means tansaction Ti issues read/write on item X)
26. Define functional dependency and explain with example. (2025)
27. What are Armstrong's axioms? (2025)
28. What is the Significance of Log-Based Recovery? (2025)
29. List and explain the desirable properties of a transaction (2025)
30. a) What is normalization? Explain about lNF, 2NF and 3NF with definition and relevant examples. (2025)
    b) What are different anomalies in designing a database? Explain each with examples.
31. Consider the relation R = {A, 8, C, D, E, F, G, H} and the set of functional dependencies F = {A→DE, B→F, AB→C, C→GH, G→H}. What is the key for R? Decompose R into 2NF and then 3NF relations. (2025)
32. a) Differentiate serial and concurrent schedules. Elaborate conflict serializability 10 with suitable example. (2025)
    b) Explain briefly the ACID properties of a transaction.

33. a) What are dirty-read and lost-update problems? Explain with the help of examples. (2025)
    b) What is the two-phase locking (2PL) protocol? How does it guarantee serializability? How strict 2PL differs from basic 2PL?

MODULE-4

1. What are the main characteristics of NOSQL systems in the areas related to data models and query languages? (2022)
2. What is a key-value database? List its major properties. (2023)